MULTIDIMENSIONAL PARAMETER DECOMPOSITION FOR MECHANISTIC INTERPRETABILITY

A PREPRINT

Knud Rasmus Sønderbek

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ABSTRACT

Some super cool abstract of all my findings

1 Introduction

Mechanistic interpretability seeks to decompose a neural network's parameters into a set of atoms, each corresponding to a concrete mechanism underlying the model's computation, with the goal of simplifying even dense and distributed networks into a simpler, human-interpretable form.

However, both the atomic structure for decomposition and the decomposition itself remain open problems. [Sha+25] Whether a decomposition recovers the computational primitives of a neural network depends on what the ground-truth decomposition is taken to be. With models consisting of independent features, the decomposition is unambiguous, with one atom per feature. If the features are dependent or co-occur, the decomposition becomes underdetermined. Decompositions that split features across multiple atoms, risk losing structural information, that gave each feature its meaning, while decompositions that group multiple features into a single atom, risk losing granularity by collapsing distinct features into a single unit.

For example, a network computing the volume of a cube can be decomposed either into components corresponding to individual side lengths, or into a single component computing the volume directly. The former corresponds to a feature-level decomposition, with one atom per input direction, exposing internal structure but leaving grouping implicit, only recoverable through co-activation patterns. The latter corresponds to a structural-level decomposition, with one atom per computation, exposing the grouping directly but collapsing per side-length information.

Recently, a formal definition of irreducibility was introduced to disambiguate cases such as the cube example. [Eng+25] Features are irreducible, and thus constitute units of decomposition, if their joint distribution cannot be decomposed into independent factors or into a mixture containing a lower-dimensional component.

In terms of the decomposition itself, recent work shifts focus from matching activations [Lea+25] to directly decomposing parameters. Guided by the minimum description length, these methods aim to recover components that are faithful to the original weights, sparse per input, and individually simple. One such approach, Stochastic Parameter Decomposition (SPD) [BBS25], factorizes networks into rank-one components with a learned per component importance function.

This work extends SPD by examining its behaviour on dependent features, in settings where the multi-dimensional superposition hypothesis posits a feature-level or a structural-level unit [1] [Eng+25]. A reformulation of the importance function is proposed, to encourage that features that are manipulated jointly are grouped together, by computing importance relationally across components.

[1] I mean feature-level, structure level is wrong under this view but it sounds good

2 Old Introduction - may be useful for related work.

Neural networks continue to achieve remarkable performance across domains ranging from medical diagnosis to autonomous systems. This increase in performance has, however, come with a corresponding increase in complexity, making these systems less interpretable to those who build and deploy them. As neural networks enter high-stakes environments, where errors carry legal, financial, or human cost, interpretability becomes a necessity. The concern is not merely technical but ethical: a fundamental principle of accountability holds that the creator of a tool can only be held responsible for its outcomes to the extent that those outcomes could have been reasonably foreseen. A system whose internal reasoning cannot be understood is, by definition, one whose failures cannot be anticipated.

Mechanistic interpretability aims to reverse-engineer the internal computations of neural networks into their simplest forms, such that they can be expressed as humanly interpretable algorithms.
Many of the current popular methods within this space, such as sparse autoencoders [Cun+23], circuit analysis, and probing, primarily operate in activation space. However, each of these methods comes with its own set of limitations. Sparse autoencoders treat features as independent directions and discard the geometric structure between them, despite growing evidence that this structure carries meaningful information (Leask et al., 2025; Mendel, 2024; Engels et al., 2024). Moreover, sparse autoencoders are trained to reconstruct activations and do not to explicitly identify the computational units that produced them. [Lea+25; Cha+25] The resulting decomposition may be functionally equivalent at a given layer, while not resembling the mechanisms that the network employs. Circuit-level methods face the deeper issue that neural networks are densely connected, meaning any given output can be equally well attributed to a multitude of causal pathways, leaving the choice of circuit underdetermined.

A more recent line of work shifts the object of decomposition from activations to parameters. More concretely, these methods aim to decompose a network's weights into a sum of simpler components, each of which corresponds to a concrete "mechanism". This reframing sidesteps several of the issues above: mechanisms are no longer constrained to specific architectural components, such as neurons or attention heads, and can instead be distributed across multiple architectural components. Moreover, they decompose the network into functional units rather than representational directions, which may further enhance their interpretability.
Recently proposed methods guided by the principle of minimum description length have shown promising results in this direction, aiming to decompose a network into parameter components that are faithful to the original weights, minimal in the number required per input, and individually as simple as possible.
One such method, Stochastic Parameter Decomposition (Bushnaq et al., 2025), decomposes networks into rank-one subcomponents. Since not all components are relevant for every input, the method learns a function that estimates each component's causal importance for a given input. However, each component's importance is calculated independently, relying only on its own activation. As noted in the original work, this is likely too restrictive for non-toy models, as it prevents each component's importance from being informed by the state of other components. This work replaces the independent importance functions with a graph neural network that determines each component's importance conditioned on its relationships to all other components, while preserving explainability through its generalised additive structure. Additionally the use of an additive importance function, generalizes the decomposition to k-dimensional components. This generalizsation is motivated by the growing evidence, that transformer models, learn features which occupy irreducible multidimensional subspaces.

[Bra+25]

3 Related work

-Open problems in mechanistic interpretability - SPD - SPD on transformers - APD - Engels et al. -Activation space decompositions - Circuits, SAE, Linear representation hypothesis Superposition in general? Strong feature hypothesis

4 Method

4.1 Background on SPD

This paper builds on the line of work of Attribution-based Parameter Decomposition (APD) [Bra+25] and Stochastic Parameter Decomposition (SPD) [BBS25], which both aim to decompose a network's parameters into a set of components that are faithful to the original weights, minimal in the number required per input, and individually as simple as possible.

To formalise this, consider a neural network $f(x, W)$ with weight matrices $W = \{W^1, \ldots, W^L\}$. A set of parameter components $\{W_c^l\}_{c=1}^C$ are said to comprise the network's mechanisms if they satisfy the following properties:

- **Faithfulness**: The parameter components should sum to the parameters of the original network.
- **Minimality**: As few parameter components as possible should be required to replicate the network's output on any given input.
- **Simplicity**: Each component should span as few weight matrices and as few ranks as possible.

Stochastic Parameter Decomposition (SPD) aims to achieve this by decomposing the weight matrices of a network into a sum of rank-one matrices, called subcomponents:

$$W^l_{i,j} \approx \sum_{c=1}^{C} \vec{U}^l_{i,c} \vec{V}^l_{j,c}, \quad \text{where} \quad \vec{u} \in \mathbb{R}^{d_{out}}, \vec{v} \in \mathbb{R}^{d_{in}} \tag{1}$$

where C is a hyperparameter, for the number of components for each layer, l indexes the layer and i, j are the hidden indices. The subcomponent count, C, may be chosen to be larger than both d_{in} and d_{out}, enabling the method to learn features in superposition [BBS25].

To enforce the properties above, three losses are introduced, which are optimised jointly. Faithfulness is enforced thorugh a mean squared error loss between the original network's weights and the the sum of subcomponents:

$$\mathcal{L}_{faithfulness} = \frac{1}{N} \sum_{l=1}^{L} \sum_{i,j} \left(W^l_{i,j} - \sum_{c=1}^{C} U^l_{i,c} V^l_{c,j} \right)^2 \tag{2}$$

where N is the total parameter count in the target model.

Optimization for simplicity and minimality is achieved through a stochastic masking procedure. Intuitively, if a subcomponent is not necessary for reconstruction, it can be sampled to zero without loss of faithfulness. A causal importance function $\Gamma : \mathcal{X} \to [0,1]^{C \times L}$ is trained to learn the lower bound of the sampling distribution. In SPD, Γ is a set of independent MLPs γ^l_c that each take a layer's activation and output a scalar importance for the corresponding subcomponent:

$$g^l_c(x) = \Gamma^l_c(x) = \sigma_H(\gamma^l_c(h^l_c(x))), \qquad h^l_c(x) = \vec{V}^{l\top}_c \vec{a}^l(x) \tag{3}$$

where σ_H is a hard sigmoid and $\vec{a}^l(x)$ is the activation at layer l. Given a stochastic mask $r^l_c \sim \mathcal{U}(0,1)$, the masked weight matrix is

$$\begin{aligned} m^l_c(x,r) &= g^l_c(x) + \big(1 - g^l_c(x)\big)\, r^l_c, \\ W'^l(x,r) &= \sum_{c=1}^{C} \vec{U}^l_{i,c}\, m^l_c(x,r)\, \vec{V}^l_{c,j}, \end{aligned} \tag{4}$$

and the training objective requires that, for all sampled masks,

$$f\big(x \,\big|\, \{W'^l(x,r)\}_{l=1}^{L}\big) \approx f\big(x \,\big|\, \{W^l\}_{l=1}^{L}\big)\,. \tag{5}$$

These masked matrices are used to compute the reconstruction loss, how well the masked weights can reconstruct the original network's output. These are computed both on a layer-wise basis and globally, as the global version can be rather noisy if all components are masked at the same time.

$$\begin{aligned} \mathcal{L}_{\text{stochastic-recon}} &= \frac{1}{S} \sum_{s=1}^{S} D\big(f\big(x \mid W'(x, r^s)\big),\ f(x \mid W)\big)\,, \\ \mathcal{L}_{\text{stochastic-recon-layer}} &= \frac{1}{LS} \sum_{l=1}^{L} \sum_{s=1}^{S} D\Big(f\big(x \mid W^1, \ldots, W'^l(x, r^{(s)}), \ldots, W^L\big),\ f(x \mid W)\Big)\,. \end{aligned} \tag{6}$$

where D is a task-specific divergence measure, such as Mean Squared Error for regression tasks or KL divergence.

The masks should also be as sparse as possible, which is enforced by penalizing the importance values toward zero:

$$\mathcal{L}_{\text{minimality}} = \sum_{l=1}^{L} \sum_{c=1}^{C} \big(g^l_c(x)\big)^p, \qquad p > 0. \tag{7}$$

The full SPD objective is a weighted sum of these terms:

$$\mathcal{L}_{\text{SPD}} = \mathcal{L}_{\text{faithfulness}} + \beta_1\, \mathcal{L}_{\text{stochastic-recon}} + \beta_2\, \mathcal{L}_{\text{stochastic-recon-layer}} + \beta_3\, \mathcal{L}_{\text{minimality}}. \tag{8}$$

4.1.1 Limitations

Sensitive to hyperparameter minimality loss,

4.2 Irreducibility and the natural unit of decomposition

The superposition hypothesis states that the hidden states can be represented as a linear combination of sparse one-dimensional features. [Elh+22; Eng+25] Concretely, the hidden state $x_{i,l}$ can be written as:

$$x_{i,l} = \sum_i v_i \, f_i(t) \tag{9}$$

where $v_i \in \mathbb{R}^d$ are pairwise δ-orthogonal directions and $f_i(t) \in \mathbb{R}$ are sparse scalar activations, with $f_i(t) = 0$ outside the domain of feature i.

Recent work has shown, that this may be insufficiently strict, [Smi24; Eng+25], as it does not capture the structure of multi-dimensional features. In particular, transformer models are shown to organise cyclical concepts in low-dimensional subspaces that do not decompose into meaningful linear components, but are instead manipulated jointly [Eng+25].

To accomodate these features, the same paper proposes a correction to the superposition hypothesis, alongside a formal definition of irreducibility that defines, when a feature is in its atomic form and when it can be decomposed into smaller units without loss of information.

A feature $f \in \mathbb{R}^{d_f}$ is reducible into features a and b if there exists an affine transformation

$$f \mapsto Rf + c \equiv \begin{pmatrix} a \\ b \end{pmatrix} \tag{10}$$

for some orthonormal $d_f \times d_f$ matrix R, and additive constant c, such that the joint distribution $p(a, b)$ satisfies one of two conditions:

- **Statistical independence**: $p(a, b) = p(a)\, p(b)$
- **Sum of disjoint distributions**: $p(a, b) = w\, p(a)\, \delta(b) + (1 - w)\, p'(a, b)$

where δ is the Dirac delta distribution and either a or b is lower dimensional in the disjoint distribution.

Under this definition, the superposition hypothesis is restated as follows:

Every hidden state $x_{i,l}(t)$ can be written as a sum of feature contributions

$$x_{i,l}(t) = \sum_i V_i \, f_i(t) \tag{11}$$

where each $f_i(t) \in \mathbb{R}^{d_{f_i}}$ is an irreducible feature with internal dimensionality $d_{f_i} \geq 1$, each $V_i \in \mathbb{R}^{d \times d_{f_i}}$ and $\|V_i^\top V_j\| < \delta$ for $i \neq j$.

The definition of $V_i \in \mathbb{R}^{d \times d_{f_i}}$ does not require columns to be linearly independent: d_{f_i} specifies an upper bound on the representational capacity of V_i, while rank$(V_i) \leq d_{fi}$, determines the effective intrinsic dimensionality required to represent the feature. For example, a model may align dependent components within a shared lower-dimensional subspace (Section 5.2.1).

4.3 Rank-k subcomponents and relational importance

In the original SPD paper, the subcomponents are constructed as rank-one matrices, through an outer product of two vectors. This formulation is however incompatible with the multi-dimensional superposition hypothesis. (Section 4.2) A decomposition into lower-dimensional subcomponents, necessarily forces features to be split across multiple subcomponents, or to be approximated through a single subcomponent at the cost of reconstruction.

To represent multi-dimensional features, the pairs of vectors are replaced with pairs of matrices.

$$W^l \approx \sum_{c=1}^{C} U_c^l {V_c^l}^\top, \qquad U_c^l \in \mathbb{R}^{d_{\text{out}} \times k}, \quad V_c^l \in \mathbb{R}^{d_{\text{in}} \times k} \tag{12}$$

Each subcomponent $U_c^l V_c^{l\top} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ has rank at most k, recovering the original SPD definition when $k = 1$. As in the multi-dimensional superposition hypothesis, k is an upper bound on each subcomponent's representational capacity; the effective rank may be lower, as it will be discussed [2] in Section 4.5.

In SPD, each subcomponent's importance is computed independently through an MLP followed by a sigmoid non-linearity, allowing multiple subcomponents to be active for a given input and thus distribute a feature across them, even when it is irreducible. Likewise, the minimality loss does not translate to the rank-k setting, where each subcomponent can represent multiple feature directions: it imposes no penalty for distributing a feature across multiple subcomponents, nor for a single subcomponent to encode parts of multiple features. (Section 4.1)

To address this, the causal importance function is redefined to maximize the conditional probability, that a given subcomponent c, is responsible for a given feature, conditioned on the activations of all other subcomponents in the same layer.

$$g_c^l(x) = p\big(c \mid \{h_{c'}^l(x)\}_{c'=1}^C\big), \quad h_c^l(x) = V_c^{l\top} \vec{a}^l(x) \in \mathbb{R}^k \tag{13}$$

where $\vec{a}^l(x)$ is the activation of layer l and C is the set of subcompoennts in layer l. The distribution is computed using a transformer block operating over subcomponents, with a softmax over their activations.

This produces a normalized assignment over subcomponents, $\sum_{c=1}^C g_c^l(x) = 1$, causing the original minimality loss to become degenerate, as each layer contributes a constant value independent of the model state. Instead, it's replaced with entropy, to pressure the assignment to concentrate on as few subcomponents as possible.

$$\mathcal{L}_{\text{minimality}} = -\frac{1}{|C|} \sum_{l=1}^{L} \sum_{c=1}^{C} g_c^l(x) \log g_c^l(x) \tag{14}$$

This pressure is complementary to both the faithfulness loss and the rank-k extension. If k is too small or the minimality coefficient is too small, the faithfulness loss dominates and forces features to split across subcomponents. Conversely, if the minimality weight is too high, the model groups features together at the cost of faithfulness.

An unfortunate limitation of this formulation is its computational cost. Computing relational importance via a transformer over subcomponents scales quadratically with the number of subcomponents compared to the linear cost of the original independent importance functions. [3]

4.4 Add the part that rank k, is unsifficient in the standard SPD, as it has no structural pressure to group things, and always that orthogonal dparts are split across components, as we only punish multiple parts being active, but we dont mind that they are split across multiple components

4.5 SVD reformulation, capacity and per feature ablation

The competition pressure introduced by the softmax has a theoretical motivation under the multi-dimensional superposition hypothesis. The hypothesis predicts that different irreducible features occupy approximately orthogonal projection subspaces; the softmax-and-attention mechanism implements a training pressure toward this orthogonality at the level of SPD's atoms. Atoms whose V matrices share input directions cannot be cleanly distinguished on inputs that activate those directions, and the softmax penalises ambiguous allocations. The optimiser's response is to push the atoms' V matrices toward disjoint subspaces, recovering the orthogonality structure the hypothesis posits.

Remember to add an experiment that shows if softmax enforce the orthogonality constraint from engels.

4.5.1 Per-dimension ablation and implicit rank selection (Hypothesis: ask lukas)

Following the extension to rank-k subcomponents, and the alternative formulation of the GNAN, the output $g_c^l \in [0,1]^k$ is a per feature importance value, and thus the masking can be applied independently to each feature. The intuition behind this is that if a mechanism captures a 1 dimensional feature, it would only need of the k dimensions to be active.
Likewise if this actually ends up working, k can be considered an upper bound or a so called "capacity" of a mechanism, it also follows nicely with the descriptive length principle found in the appendix of APD.

To determine the effective rank of each subcomponent, the stochastic masking becomes:

$$[m_c^l(x,r)]_j = [g_c^l(x)]_j + (1 - [g_c^l(x)]_j) r_{c,j}^l, \qquad r_{c,j}^l \sim U(0,1) \tag{15}$$

[2]not a discussion
[3]Play with attention approximations

for each j in $1, \ldots, k$
Notably this treats each feature independently, which may not be desirable, as such we could modify it ever so slightly:

$$[m^l_c(x,r)]_j = [g^l_c(x)][g^l_c(x)]_j + (1 - [g^l_c(x)][g^l_c(x)]_j) r^l_{c,j}, \qquad r^l_{c,j} \sim U(0,1) \tag{16}$$

The signal becomes an importance of two factors, if the component is important and both features are important, it remains active, if only one feature is important, the other will be pushed to zero

The masked weight matrix becomes:

$$W'^l(x,r) = \sum_{c=1}^{C} U^l_c \operatorname{diag}\big(\mathbf{m}^l_c(x,r)\big) V^{l\top}_c \tag{17}$$

Where the diagonal matrix is the per-feature mask for subcomponent c.

The importance minimality loss penalises all k dimensions of all subcomponents toward zero:

$$\mathcal{L}_{\text{import}} = \sum_{l=1}^{L} \sum_{c=1}^{C} \sum_{j=1}^{k} \big|[g^l_c(x)]_j\big|^p \tag{18}$$

This follows the same logic by which SPD selects the number of active subcomponents: the minimality loss drives unused components toward zero, and only those that are genuinely required to reconstruct the target model's output survive.

This makes k a soft upper bound rather than a sensitive hyperparameter. In practice, k can be set conservatively large and the effective rank of each subcomponent is learned from the data.

5 Experiments

5.1 Reproducing SPD on independent features

5.1.1 Sensitivity to hyperparameters

5.2 TMS with tied weights and synced inputs

To examine how SPD and the softmax-and-attention extension handle irreducible features, we reuse the model of superposition introduced in Section 5.1. The architecture is a single-hidden-layer autoencoder with tied weights:

$$f(x) = W^\top \sigma(Wx + b), \qquad W \in \mathbb{R}^{d_{\text{hidden}} \times d_{\text{features}}}, \quad d_{\text{features}} \gg d_{\text{hidden}} \tag{19}$$

The model is trained to reconstruct sparse feature activations under an L_2 loss. The input distribution differs from Section 5.1 in that features are tied into pairs. Concretely, $d_{\text{features}} = 40$, $d_{\text{hidden}} = 10$, and the features are partitioned into 20 pairs $\{f_{2i-1}, f_{2i}\}_{i=1}^{20}$. Each pair is zero with probability $1 - p$, and with probability p both features are independently drawn from $\text{Uniform}(\epsilon, 1)$.

By construction, the joint distribution of each pair satisfies neither of the reducibility conditions defined in Section 4.2: the activation of one feature in a pair fully determines the activation of the other, and the joint distribution has no mass where one feature is zero while the other is active. By [Eng+25], each pair is therefore an irreducible two-dimensional feature.

Since $d_{\text{features}} \gg d_{\text{hidden}}$, feature representations are necessarily compressed through superposition. In the learned encoder, this results in near-collinearity within feature pairs and approximate orthogonality across pairs.

Group	Mean	Std Dev	Range
Within-group	0.999	0.010	$[0.997, 1.000]$
Across-group	0.053	0.223	$[-1.000, 0.038]$

Table 1: Cosine similarity between encoder columns $W_{:,i}$ and $W_{:,j}$, partitioned by group membership.

Under the multi-dimensional superposition hypothesis (Section 4.2), the formal dimensionality $d_{fi} = 2$, which provides an upper bound on the rank required to represent each feature. The observed near-collinearity within pairs suggests that this representation essentially collapses to a one-dimensional subspace.

To quantify this, the effective rank of each pair's submatrix, $W_{:,2i-1:2i} \in \mathbb{R}^{d_{\text{hidden}} \times 2}$ is computed using the entropy-based definition of [RV07].

$$r_{\text{eff}}(W_g) = \exp\left(-\sum_{k=1}^{Q} p_k \log p_k \right), \qquad p_k = \frac{\sigma_k^2}{\sum_{j=1}^{Q} \sigma_j^2} \tag{20}$$

where σ_k are the singular values of W_g and $Q = |g|$. The quantity lies in $[1, |g|]$ and equals 1 when the group is exactly rank-one. To gauge the required value of k for the rank-k extension, the maximum effective rank across pairs is examined:

$$k = \max_i, \; r_{\text{eff}}(W_{:,2i-1:2i}), \tag{21}$$

In this case the max is essentially 1 across all pairs, suggesting that $k = 1$ is sufficient to capture these features without loss of information. (Table 2).

Mean	Std Dev	Range
1.0003	0.01	$[1.0, 1.05]$

Table 2: Effective rank of paired-feature encoder blocks. Values near 1 indicate collapse to a single direction.

5.2.1 Comparison of vanilla SPD and the softmax-and-attention extension

Both decompositions are trained with $C = 150$ subcomponents and the hyperparameters proposed in the original SPD paper, with the exception of the minimality loss, which is set to x for vanilla SPD and y for the softmax-and-attention extension. We use $k = 1$, since each pair was shown to be representable along a single direction in hidden space. The ground truth decomposition is one where each pair is captured by a single subcomponent.

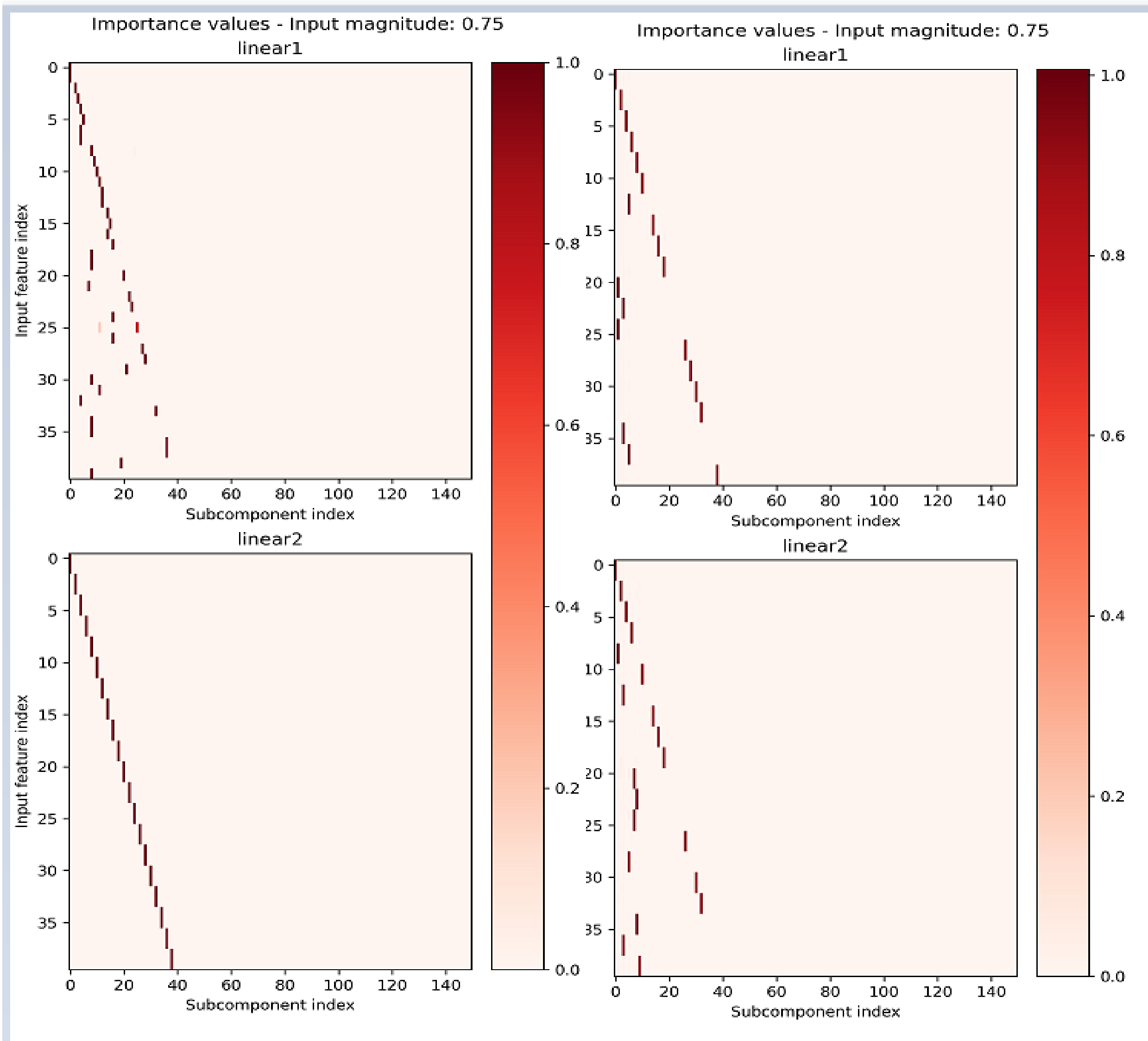

Figure 1: Left: Attention and softmax only at like 15k steps vs 40k steps, Right: Vanilla spd.

Text explaining what we see, this will change so wait for now. -Something about if both decompositions are faithful, and if they are faithful to one of the two hypotheses, if they are delta orthogonal, and if the importance values are interpretable.

To verify a successful decomposition, each matrix V_i should also be δ-orthogonal to the others.

The multi-dimensional superposition hypothesis predicts that the projection matrices for different irreducible features are pairwise δ-orthogonal. At the parameter level, the recovered V_c matrices are the operational counterpart, and the prediction translates to the requirement that recovered atoms have small pairwise overlap in their input-side directions.

For each method, we compute the pairwise cosine similarity [4]

$$M_{cc'} = \frac{V_c^\top V_{c'}}{\|V_c\| \|V_{c'}\|} \tag{22}$$

compared against the across-group cosine similarity of the trained model's encoder $(0.053, 1)$, the softmax-and-attention shows ”something” and the baseline model finds that ”something else”.

[4]Maybe even UV combined? Only read or write?

here we have four nice plots, overall alligment, across V's, across U's, entire thing? Reference to original model and maybe a table?

Remarks for this section Intra cluster cosine has a -1. Two features are perfectly alligned to a factor of -1, can this cause issues with decomposition? Same component could capture both features, if they read from the same input. However they shouldnt as its distinctly different feature indices.

5.3 Extending to Rank-k: When Higher-Dimensional Structure Matters

The rank-k extension is required in settings where feature representations do not collapse to a single direction. This is illustrated by a model trained to compute the volume of an n-dimensional simplex.

In this experiment, consider a model trained to compute the volume of an n-dimensional simplex:

$$f(x) = W_1\sigma(W_2x), \quad W_2 \in \mathbb{R}^{k \cdot n(n+1) \times d_{hidden}}, \quad W_1 \in \mathbb{R}^{d_{hidden} \times k} \tag{23}$$

Here, k denotes the number of simplices and n the dimensionality of each simplex, which has n+1 vertices.

As in Section 5.2, the k simplices are sampled independently with fixed probability, such that either all n(n+1) features of a simplex are active or none are. The resulting feature groups are therefore statistically dependent, and their joint distribution assigns negligible probability to partial activation patterns. By the definition of irreducibility in Section 4.2, each simplex constitutes an irreducible feature with formal dimensionality $d_{fi} = n(n+1)$.

For this experiment, the model is trained with $n = 2$ and $k = 4$, yielding four triangles $\mathbb{R}^2$ and a total of 24 input features. The ground-truth decomposition therefore consists of four subcomponents, each responding to 6 adjacent features. We note that the effective rank and cosine similarity here imply a decomposition should require at least $k = 2$. **[Placeholder: cosine similarity results]**

[Placeholder: effective rank results, showing $r_{eff} > 1$] Firstly, we demonstrate that the original SPD formulation surprisingly yields the best clustered reconstruction when $k = 1$:

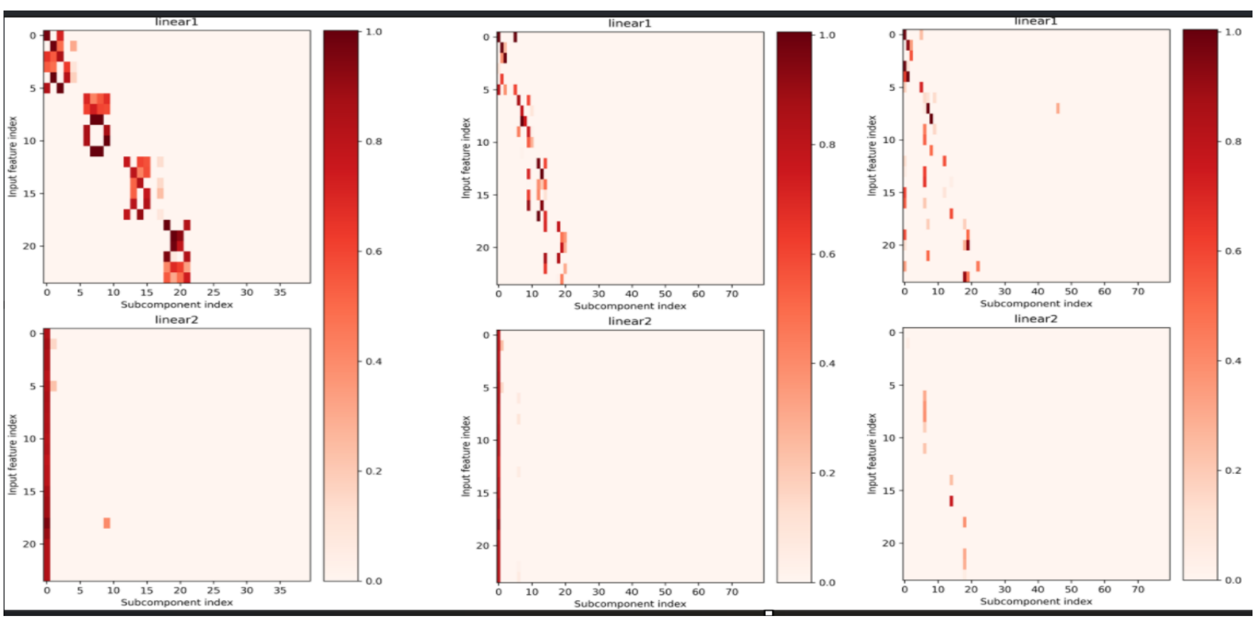

Figure 2: SPD decomposed with k=1,2,3 from left to right.

The reconstruction clearly yields 4 subcomponents per group, that only respond to the 6 features of a single simplex, however these subcomponents are seperated even though they are irreducible.

We note that we werent able to find a reconstruction at all when the minimality coefficient was outside of $[1e{-}4, 5e{-}4]$.

The softmax and attention extension, on the other hand, shows the expected behaviour, with $k = 1$, the reconstruction neccessitates that each simplex is split into multiple subcomponents, here each subcomponent captures a single feature. The extension to $k = 2$ allows the model to capture each simplex as a single unit, with all 6 features of a simplex captured by a single subcomponent. However, it does collapse 2 simplices into a single subcomponent. The extension to $k = 3$ does not yield any improvement, suggesting that the model has found a faithful reconstruction at $k = 2$.

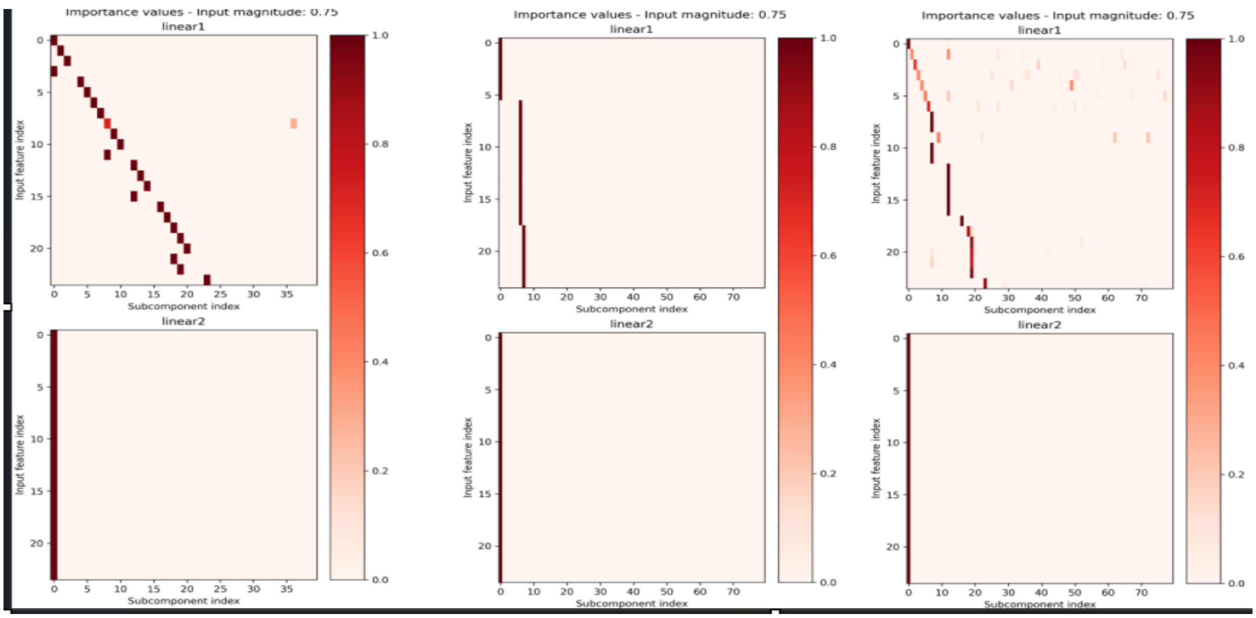

Figure 3: Softmax-and-Attention decomposed with k=1,2,3 from left to right.

While the softmax approach does not find a perfect reconstruction either, it clearly demonstrates how the extension from rank 1 to rank 2 allows a non diagonal representation of the features. Had it been perfect, then the middle plot would have shown 4 components instead of the collapsed 2.

Same plots i was going to do for the TMS experiment, perhaps this gets its own section, we compare how input spaces are alligned over training, does the softmax encourage δ orthogonality better?

5.4 Checking for orthogonality throughout training with softmax and attention / compare to baseline spd.

5.5 Toy model of learning gates

Remark: Im not sure we can sample binary values, double check the distribution, i think it requires a continuous thing with b not fixed? Could always train an addition model, but they already did that in the original paper? Can we just use that one to prove the claim?

To test the case of enges, where the probability distribution is in fact reducable. We could train it to learn logic gates i.e 1+0 = 0, 1 + 1 = 1, 0 + 0 = 0, 0 + 1 = 1, in such a view each could occur independently ,but nlywhen both are active, the output is active, we then sample independently from the two features and see if it compacts them in a single unit or not.

The concept here is clear: The ideal decomposition here is one that produces independent features, that can co occur, however the atomic solution is independence.

We also need to probe for a paired result here. The ground truth should show part ownership across 2 components.

bool A, bool B are independnet their joint result shoul be shared ownership.

5.6 Extending the rank to be an upperbound through per feature ablation

5.7 Irreducibility as the target of decomposition

6 Discussion

7 Acknowledgements

8 References

9 Other things to consider?

Minimal descriptive length?

9.1 That some features are multidimensional

Think back to SAE, that learn a dictionary of features, then find hte linear combination of those features, however that this fails for cyclical features, as its cheaper to learn a single feature, than learn a direction and a periodic function. Now for your experiment, the dream scenario is that we allowe a multi dimensional component, and that this can learn the cyclical feature, i.e a weekly cycle or modular arithmatic either do this or extend TMS to be multi-dimensional, show if it works better with ur gnn or not, likewise we need to show that through individual masking of features, we can teach a component to kill one of its features, if it is better explained by a rank 1 component than rank 2.

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9.1.1 GNAN-based importance functions

In SPD, each importance function γ_c^l is an independent MLP that receives only its own inner activation. Intuitively, however, causal importance is relational: whether a subcomponent can be ablated without affecting the output may well depend on which other subcomponents are active alongside it.

To address this, the independent MLPs are replaced with a Graph Neural Additive Network (GNAN) that can relate each subcomponent to its neighbourhood.

The C subcomponents are treated as nodes in a graph, with node features given by their inner activation vectors $\mathbf{h}_c^l(x) \in \mathbb{R}^k$. The GNAN updates each node's representation through an additive neighbourhood aggregation. For node i in layer l and feature dimension k [5]:

$$[\mathbf{z}_i]_k = \sum_{j=1}^{N} \frac{1}{|\mathcal{L}(j)|} \cdot \rho\bigg(\frac{1}{1+\mathrm{dist}(j,i)},\ \cos(j,i)\bigg) \cdot f_k([\mathbf{h}_j]_k) \tag{24}$$

OBS: Implementation ill present today is this:

$$\mathbf{z}_i = \sum_{j=1}^{N} \rho\,(\text{layer distance}, \text{cosine similarity}) \cdot f_k\,([\mathbf{h}_j]_k) \tag{25}$$

where f_k is a learned function applied to the k-th inner activation of each neighbouring node j; $|\mathcal{L}(j)|$ is the number of nodes in the same layer as j, which normalises the contribution of each layer; $\mathrm{dist}(j,i)$ is the layer distance between nodes j and i[6]; $\cos(j,i)$ is the cosine similarity between their parameter vectors; and ρ is a learned function that maps both the distance kernel and the cosine similarity to a scalar weight.

z_i becomes a vector in $\mathbb{R}^k$, which fits perfectly with the current activation vectors.

A separate MLP is trained to combine the aggregated neighbourhood information and the node's own activation into an importance value in $[0,1]$: Producing a single importance value per subcomponent and passing it through a leaky hard sigmoid, following SPD [BBS25]:

$$g_c^l = \sigma_H(AGG(\mathbf{z}_i^l, \mathbf{h}_i^l)) \tag{26}$$

The self-connection through f_{self} (or just the raw activation) ensures that each node retains its own identity after aggregation, mitigating the oversmoothing that can occur in GNNs. {Clever cite here, i.e gnn book}

Alternative formulation, that would work with the next section

After the GNAN update, the per-dimension importance values are computed from the updated representation and passed through a leaky hard sigmoid, following SPD [BBS25]:

$$\mathbf{g}_c^l(x) = \sigma_H\big(\mathbf{h}_c^l(x)\big) \in [0,1]^k \tag{27}$$

in other words, the importance function becomes multidimensional, and each column can be masked independently.

Remarks:
For now the edge set construction is left as the fully connected graph, which remains reasonably computationally expensive for small models. However it could very easily be made more efficient, for other architectures, or by considering only the k-hop neighbourhood of each node, or only considering prior neighbours or later neighbourhoods. Likewise we could store both distances as sparse structures as they are symmetric. $\cos(j,i) = \cos(i,j)$
Remark 2:
The choice of edge set construction and distance function is a very important design choice, e.g using the layer distance on a fully connected graph encodes the same signal for all nods in a layer: For nodes i_1, i_2 in the same layer l:

$$\forall\, j: \quad \mathrm{dist}(j,i_1) = \mathrm{dist}(j,i_2) = |k_j - l| \tag{28}$$

and therefore

$$[\mathbf{h}_{i_1}]_k = \sum_{j=1}^{N} \frac{1}{\#\mathrm{dist}_{i_1}(j,i_1)} \cdot \rho\bigg(\frac{1}{1+|k_j-l|}\bigg) \cdot f_k([\mathbf{x}_j]_k) = [\mathbf{h}_{i_2}]_k \tag{29}$$

since no term in the sum depends on the identity of i within layer l, only on the layer index l itself.
Additionally this produces a counterproductive signal, the gnan tries to boost or suppress all nodes in a layer together, instead of learning to differentiate between them.

Remark 3:
The choice of distance should probably be reconsidered in general, I have a feeling that the function needs to be from an asymmetric metric space, i.e

$$dist(i,j) \neq dist(j,i) \tag{30}$$

[5]I am very open to alternative equations, i.e perhaps we delete the layer distance inside rho, however it seemed necessary to use cosine similarity, as otherwise the signal was useless as described in some earlier section

[6]The layer distance, could later be signed to differentiate between previous and future layers

Atleast if the intution is if component A in layer k, is highly active, perhaps component B in layer k+1 should likely be inactive, as the other component needs to handle this feature, thus the distance from A to B should be different, as otherwise both are boosted or suppressed together.

Remark 4:
The current experiments are more sensitive to hyperparameters, than the original code seems to be, if the minimality pressure is too low, then all components are active, and doesn't "kill" any of them

Remark 5:
I haven't been able to find any "benefit" in setting k higher, this is ofcourse because hte ground truth of all the models we have are rank 1 independent features, we need to find a scenario where the dependence and or rank k is actually necessary:

My ideas

- Sample points from the unit circle, i.e $\cos(\theta)^2 + \sin(\theta)^2 = 1$ then

  $$x = \sqrt{1 - \sin(\theta)^2} \text{ or } - \sqrt{1 - \sin(\theta)^2}$$

  i.e sometihng can only be expressed with a rank 2 component?
- we could look for circuits, the goal is then no longer to find N independent features, but suppose we have input groups [0,1,2], [2,3,4], [3,4,5] then we want one component that activates for the first group, one for the second and one for the third

Remark 6:
I added the gnan plots to wandB remember to show them

Remark 7:
The per feature construction with only 1 feature is rather boring, and does not utilize the expressiveness at all.

Remark 8:
I dont think its a code problem, as i can reproduce equivalent results to their findings. Im just not able to beat them. We need to discuss experiments where the k dimensionality and dependence actually matters.

10 Notes

Putting the minimality lsos entropy higher, encourages diagonals? Instead of capturing two features in the same component.

Increasing minimality loss encourages feature grouping.